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Enhanced Ransomware Detection using a Hybrid Quantum-Classical Convolutional Neural Network with Boosted Sooty Tern Optimization

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***Abstract*— Ransomware attacks threaten organizations by encrypting files or locking systems and keeping them inaccessible unless a ransom is paid. Early detection of ransomware attacks helps organizations avoid financial losses, theft of data, and harm to their reputations. This early detection will also reduce the risks related to malicious activities. Detection by signature-based or heuristic approaches tends to be traditional and fails when evolvement in ransomware attacks is considered. Thus, a need for innovative techniques of deep learning and optimization algorithms can be observed. The method proposed in this study focuses on developing a novel multi-stage framework for detecting ransomware attacks. Firstly, correlation coefficient-based min-max normalization (CCMMN) for data preprocessing is used since it makes the data elements more uniform and makes the detection system more efficient. Feature selection is then carried out and optimized using the hybrid Osprey optimization algorithm with an enhancing sparrow search algorithm (OOA-ESSA) to arrive at the most relevant feature for classification. Last, this paper applies a hybrid quantum-classical convolutional neural network with boosted sooty tern optimization (HQCCNN-BSTO) in ransomware detection. The classic model is replaced by a quantum and hybrid classical neural network that may be complemented by an optimization algorithm to increase the speed and efficiency of the detection. The method that has been proposed here offers a flexible and efficient approach to the problems associated with the detection of ransomware threats in the practical environment. Thus, high effectiveness is proved by the 99% accuracy rate of the suggested method. This exceptional outcome proves the robustness of the approach and its ability to adapt to various patterns of dependence and data changes.**

Keywords*— *Boosted Sooty Tern Optimization, Correlation Coefficient-Based Min-Max Normalization, Enhancing Sparrow Search Algorithm, Osprey Optimization Algorithm, Ransomware.

I. INTRODUCTION

Malware known as ransomware encrypts digital data or alters system login to lock its victim's resources. After that, it requests a ransom from the victim to release the resources. This kind of malware gained widespread recognition in the field of cybersecurity after WannaCry launched a global cyberattack in 2017 that infected over 200,000 systems in 150 different countries [1]. Ransomware is still considered to be one of the most dangerous types of malware, and cybersecurity experts keep an eye out for it. Its influence on monetary loss has grown despite a decrease in its infection rate. This is because targeted corporations, not the general public, are being attacked by thieves using ransomware. Consequently, a significantly bigger ransom is also demanded.

Ransomware is a type of virus that locks down a target computer or encrypts its files, making the device and its contents unusable. Ransomware attacks are used by cybercriminals to demand money from their victims [2-3]. Ransomware attacks are one way that nation-state actors can damage their enemies' vital infrastructure. To force the victim to pay a ransom or sell the data on the dark web, these assaults are frequently coupled with the exfiltration of the victims' data. Approximately 70% of organizations experienced ransomware attacks in 2022. By 2031, ransomware is predicted to strike a company, customer, or gadget every two seconds, up from every eleven seconds in 2021. The cost of the damage was $20 billion in 2021 and is probably going to be more than $265 billion by 2031 [4].

The prevalence of ransomware has increased dramatically worldwide in recent years. Ransomware is characterized by its rapid spread, broad influence, and significant damage. Particularly when ransomware attacks target critical information systems and infrastructure, the firms will suffer not only financial losses but also data loss, decreased productivity, regular business interruption, and other consequences. It is crucial to identify ransomware attacks, particularly those that are novel and unidentified [5]. Deep Packet Inspection (DPI) and other techniques have historically been used to identify ransomware to a certain degree. However, attackers start using encryption technology to evade detection when security protocols like SSL or TLS appear.

Due to this, even though ransomware is still in existence, it changes its modus operandi, making it very difficult to detect and prevent. Modern hackers use different levels of encoding and take advantage of computer applications and operating systems, while traditional security methods prove to be insufficient. In addition, the recent trend of ransomware-as-a-service (RaaS) means that anyone can access the tools needed for a cyberattack, regardless of technical skills. Apart from financial losses, these attacks are likely to compromise organizational data and cause reputational loss, penalties, and regulatory fines for organizations that are targeted. Higher rates of these attacks are following high-value targets like health care, finance, and critical infrastructure, emphasizing the need for better defensive and preventive measures [6-7]. With emerging threats, organizations have to add one layer after another of protection that utilizes old early warning systems

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with deep learning, behavioral analysis, and real-time threat intelligence.

Novelty and Contribution

- Preprocessing with correlation coefficients with min-max normalization guarantees ideal data scaling, enhancing ransomware detection's precision and effectiveness in a variety of settings.
- The most pertinent features are efficiently chosen by the hybrid Osprey Optimization Algorithm with Enhancing Sparrow Search Algorithm, which lowers dimensionality and improves the detection model's performance.
- The proposed method combines the computational benefits of quantum and classical convolutional neural networks, improving the model's capacity to identify intricate ransomware patterns more quickly and precisely.
- The optimization of the deep learning model is further improved by the application of Boosted Sooty Tern Optimization (BSTO), which improves detection capabilities and offers a notable improvement in ransomware classification.
- The multi-phase approach of the suggested solution provides a very flexible and robust detection mechanism, guaranteeing efficient detection and defense against changing ransomware attacks in dynamic, real-world settings.

The outstanding of Section 2 examines the literature, Section 3 suggests methods, Section 4 displays the results and discussion, and the conclusion is given in Section 5.

II. LITERATURE SURVEY

Singh et al. (2023) [8] have presented a new RANSOMNET+ that is a state-of-the-art hybrid model that classifies ransomware attacks by integrating CNNs with pre-trained transformers. It also revealed that interpretability analysis, as well as graphical representation, positively impacted its decision-making. RANSOMNET+ is reliable for CSPs and researchers, and it signifies a significant improvement in ransomware studies and cloud security. The limitation of the suggested method is that it only operates for identified strands, and generalization to unseen ransomware forms is restricted.

Ashwini and Nagasundara (2024) [9] have developed a procedure to detect and classify ransomware threats with the aid of the enhanced dual-vision transformer and the Mantis Search Split Attention Network. In the model, feature selection procedures are accelerated by the Hybrid Termite Alate City Council's Evolution Optimization technique and pre-processing HTCEO. The resulting model holds high achievement in accuracy, F1 score, recall, detection rate, Matthews's correlation coefficient, and precision. Showing how effective the model is in the protection of cybersecurity against ransomware threats. One disadvantage of the suggested method is the high computing cost of real-time detection.

In 2023 Thummapudi et al. [10] presented a fairly efficient and most helpful method to detect ransomware in a virtual machine. It removes the burden of observing and analyzing each process repeatedly; instead, it employs a machine learning classifier to collect data for certain processor and disk I/O operations. The random forest classifier performed better than all other classifiers, giving results with an accuracy of 98% in the worst-case scenario, identifying the ransomware within 400 ms across the six user loads and 22 ransomware incidences. The only drawback associated with the suggested method is on I/O events on the CPU and disk.

Moreira et al. (2023) [11] have shown an efficient approach to ransomware detection that employs an improved Xception CNN algorithm based on portable executable headers. Crypto-ransomware is a special kind of malware that threatens computer systems, and its usage has increased due to the COVID-19 pandemic. A method called Xception ColSeq converts the PE header files into color images and then uses an Xception Convolutional Neural Network model to classify these images to identify ransomware through static analysis. It can be applied to enhancing the efficiency of existing approaches in parallel with the protection of computer systems from cyber threats. The suggested method has a disadvantage, which is its applicability to Portable Executable (PE) files only.

A. Problem Statement

Currently, it is challenging to identify ransomware attacks because of poor preprocessing of these attacks, wrong choice of features, low accuracy, and high computational complexity. This study proposes a performance improvement strategy that incorporates advanced techniques. HQCCNN is applied for classification purposes; similarly, hybrid OOA-ESSA is applied for the selection of features and correlation coefficient-based min-max normalization for preprocessing. The study also incorporates the BSTO for parameters of optimality and achieves better results in the identification of attack activity with low computation complexity. This goes a long way in ensuring the improvement of the underlying problems with detecting ransomware attacks.

III. PROPOSED METHODOLOGY

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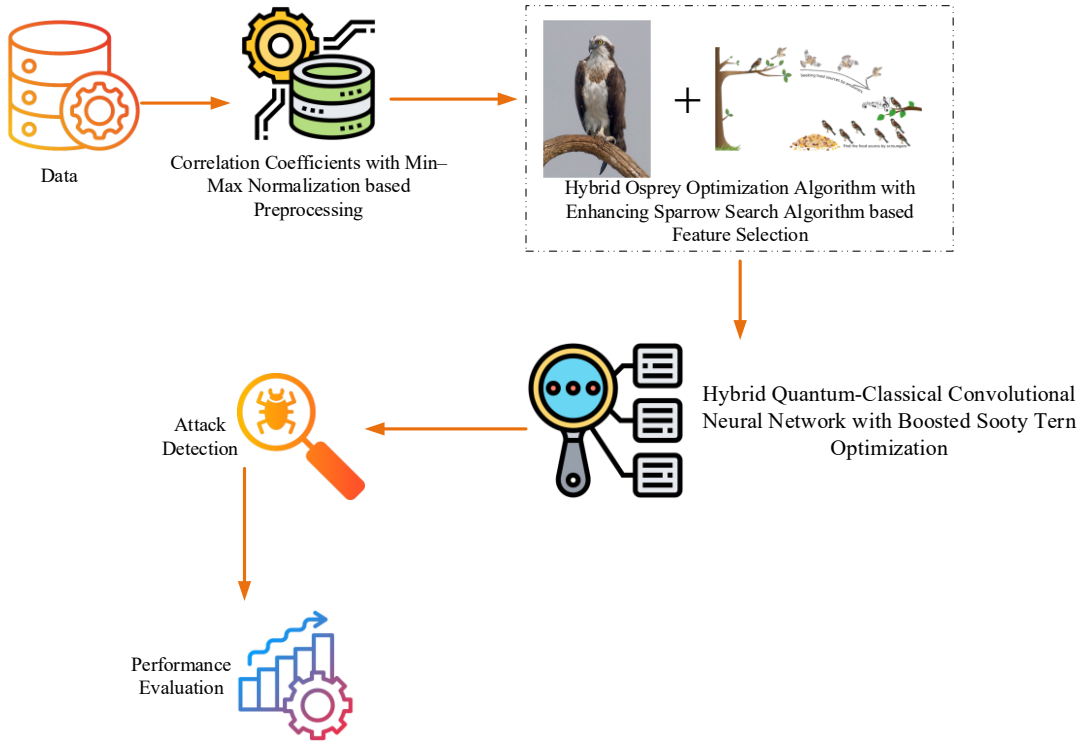

Figure 1: Illustration of the proposed methodology

The proposed process for the detection and evaluation of a ransomware attack using advanced optimization and neural network techniques is provided in the Figure 1 flowchart. This starts with the acquisition of data, which, after the acquisition, are preprocessed via correlation coefficients with data normalized through Min-Max normalization. This preprocessed data is put through pass-through feature selection using a hybrid Osprey optimization algorithm integrated with an enhancing sparrow search algorithm. The selected features are fed into a hybrid quantum-classical convolutional neural network, in which boosted sooty tern optimization is used for optimizing the attack detection task. Finally, the performance of the attack detection system is analyzed.

A. Dataset

The proposed approach is conducted on the CICIDS2017 dataset. The diversity of the attacks presented in the dataset facilitates a comprehensive evaluation of how well the method performs to detect and classify various types of attacks. The datasets are preprocessed with min-max normalization on correlation coefficients for maintaining uniform scaling across features. The method minimizes bias introduced due to varying scales of different features that impact subsequent classification performance.

B. Correlation coefficients with min-max normalization-based preprocessing (CCMMN)

The input data are sent into preprocessing to scale and normalize the outputs, ensuring consistency and relevance for accurate analysis. To ascertain the correlation between the characteristics, the suggested approach may make use of the Pearson correlation coefficient (PCC) (Equation 1) and Min-Max normalization (MMN) (Equation 2) [12]. With NEW_MN set to 0 and NEW_MX set to 1, the revised range of feature values is shown by NEW_MN and NEW_MX in the setting of Min-Max scaling (0, 1). Equation (3) is utilized to generate the updated weighted value.

$$k = \frac{\sum_{i=1}^{n} (s_i - \bar{s})(f_i - \bar{f})}{\sqrt{\sum_{i=1}^{n} (s_i - \bar{s})^2}\ \sqrt{\sum_{i=1}^{n} (f_i - \bar{f})^2}} \quad (1).$$

$$mmn, s'_{i,m} = \frac{s_{i,m} - MN(s_i)}{MX(s_i) - MN(s_i)}(NEW_mx - NEW_mn) + NEW_mn \quad (2).$$

$$\begin{aligned} &\text{New weighted value, } h' \\ &= \left(\frac{h_{i,j} - ni_j}{ns_j - ni_j}(NEW_MX - NEW_MN) + NEW_MN\right) \\ &+ C\left(\frac{h_{i,j} - ni_j}{ns_j - ni_j}(NEW_MX - NEW_MN) + NEW_MN\right) \cdot Corr(H_j, H_{target}) \end{aligned} \quad (3).$$

Where h' denotes the updated weighted value, $h_{i,j}$ is every initial value, C represents the value to modify the new range of values, $Corr(H_j, H_{target})$ utilizing PCC, the correlation coefficient between column j and columns label, ni_j denotes the column j 's minimum value, ns_j denotes the column j 's maximum value, NEW_mn represents the new minimal value is zero, and NEW_mx is the updated maximum value is one.

The first step in normalizing the dataset is to scale it to a specific range using Min-Max normalization. To examine the correlations between variables, the correlation coefficients are calculated following normalization. Using the Correlation Coefficients with Min–Max Normalization (CCMMN) method, which weights the data based on the correlation analysis conducted in the preceding steps and is best suited for the subsequent feature selection and classification, the final step is the processed data generated in this work. Preprocessing techniques like min-max

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normalization with correlation coefficients are used to make sure the data are effectively scaled and standardized, enhancing their consistency and analytical value. To choose the pertinent features, the preprocessed data is then fed into feature selection.

C. Hybrid Osprey Optimization Algorithm with Enhancing Sparrow Search Algorithm-based Feature Selection (OOA-ESSA)

The relevant features are chosen from the preprocessed data using feature selection. To improve classification accuracy with fewer features and less time complexity, the Hybrid Osprey Optimization Algorithm with Enhancing Sparrow Search Algorithm (OOA-ESSA) seeks to select relevant characteristics from the dataset. By combining the exploration features of the Osprey Optimization Algorithm with the exploitation benefits of the Enhancing Sparrow Search Algorithm, the technique ensures a local exploiter and a global scout, both of which are critical in optimization procedures. This procedure makes the forecasting model suitable for ransomware attack detection by improving its classification efficacy, reducing the data set's dimensions, and improving accuracy.

a. Osprey Optimization Algorithm (OOA) [13]

Ospreys, sometimes known as fish, river, or sea hawks, are predatory birds that hunt fish throughout the day. Its dimensions are 50-66 cm in length, 127-180 cm in wingspan, and 0.9-2.1 kg in weight. Fish make up 99 percent of the diet of ospreys, which are piscivorous birds. They typically catch live fish that weigh between 150 and 300 g and are 25 to 35 cm long. Hovering between 10 and 40 meters above the ocean, ospreys have eyesight that allows them to recognize objects underwater. When they get close to the fish, they dip their feet into the water and dive in to capture it. They capture the fish and carry it to a neighboring rock, where they eat it.

b. Enhancing Sparrow Search Algorithm (ESSA) [14]

The way sparrows naturally balance exploration and exploitation while looking for food served as the model for ESSA. SSA may navigate through more complex problem spaces than are typically possible with ordinary search algorithms thanks to techniques like hazard-aware transfer for exploitation and uniformity-diversification for exploration. The dynamic evolutionary approach's flexibility is further improved by allowing it to modify the search strategy based on the type of problem. By striking an adaptive balance between exploration and exploitation, SSA can converge to optimal solutions and successfully avoid local optima.

Initialization

Agents continuously search the search space and choose features for detection tasks in the hybrid OOA-ESSA initialization process. Each agent is assigned a binary vector, and their locations are randomly selected from a specified search area to effectively explore and select pertinent qualities for accurate detection tasks.

$$K_{i,j} = lb_j + rand_1 \cdot (ub_j - lb_j) \cdot (1 - \exp(-\alpha \cdot rand_2)) + \gamma \cdot \sin\ (rand_3 \cdot \pi) \quad (4).$$

Where $K_{i,j}$ denotes the position of the $i-th$ solution in the $j-th$ dimension, $rand_1$, $rand_2$ and $rand_3$ denotes the uniformly sampled random values from [0,1] that reflect the stochastic dynamics of osprey and sparrow. α represents a control parameter that counteracts the effects of sinusoidal behavior (from OOA) and exponential exploration (from ESSA). A scaling factor γ is used to regulate the effect of sinusoidal oscillations on local diversification.

Fitness Function

The fitness function for the proposed hybrid OOA-ESSA optimization technique is represented by Equation (5).

$$F_{function} = \beta \cdot (1 - acc) + \alpha . \frac{|I|}{|G|} \quad (5).$$

Where $F_{function}$ denotes fitness value, acc indicates the categorization accuracy attained using the selected attributes, I indicates the number of selected features in the subset, G represents all of the characteristics in the dataset, β and α balancing the trade-off between classification performance and the quantity of chosen attributes by weighting factors.

Exploration

When submerged, ospreys use their keen vision and powerful hunting skills to find fish. After locating the fish, they hunt and attack it by diving beneath it. OOA is better able to explore the optimal zone and steer clear of local optima by mimicking an osprey attack on fish, which drastically alters the osprey's position inside the search area. Equation (6) establishes the number of fish that each osprey has.

$$KL_x = \{M_h \,|\, h \in \{1,2,..., N\} \wedge K_h < K_x\} \cup \{M_{Best}\} \quad (6).$$

Where M_{Best} is the optimal candidate solution and KL_x is the collection of fish locations for the $x-th$ osprey.

Exploitation

Sparrows need to be able to sense danger and move to safer areas to forage. These individuals help the solution space find the finest solutions. The danger-aware Transferring Strategy (HATS) is optimized for danger perceivers. The distance between the best and worst locations is divided to find the moving distance. The weighted sine and cosine approach, which is proposed to lessen the likelihood of reaching a local optimal state, is based on the growth function model.

The combined technique is used for the hazard-perceivers' location update and is given in Equation (7) by combining the sine and cosine algorithm with the regulating factor (rf) and weight factor (wf).

$$Posi_{a,k}^{g+1} = \begin{cases} wf \cdot * Posi_{Best}^{g} + rf \cdot * Sin(r_2) \cdot * \left| r_3 \cdot * Posi_{a,k}^{g} - Posi_{Best}^{g} \right| & Fitness_a \neq Fitness_b \\ wf \cdot * Posi_{a,k}^{g} + rf \cdot * Cos(r_2) \cdot * \left| r_3 \cdot * Posi_{a,k}^{g} - Posi_{Worst}^{g} \right| & Fitness_a = Fitness_b \end{cases} \quad (7).$$

Where r_2 and r_3 are random values between 0 and 2 pa and 0 and 2, respectively.

Termination

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The hybrid optimization's termination condition is established by increasing the number of iterations $(e = e+1)$ following each step. The hybrid OOA and ESSA feature selection method efficiently explores and improves feature subsets by combining the best aspects of each methodology. The most crucial features of the dataset are chosen using this technique to improve efficiency and optimize the model. The selected features are then grouped into categories.

D. Hybrid Quantum-Classical Convolutional Neural Network with Boosted Sooty Tern Optimization (HQCCNN-BSTO)

The chosen features are then fed into the ransomware attack and classified using a hybrid quantum-classical convolutional neural network (HQCCNN). The QCCNN architecture is made up of a fully connected layer, pooling layers, and interlaced convolutional layers. The convolution layer, the most computationally demanding stage, uses a feature map to extract attributes from the input data. The data's dimensionality is reduced, and over-fitting is prevented with a pooling layer. Connecting input and output neurons is a single filter that selects windows according to input and window dimensions. One filter that associates input windows with output neurons uses an array F as a parameter. A two-dimensional array K with window sizes selected has the first window in the upper left corner [15]. The mapping is done using a linear function.

$$K_{1:p,1:q} \rightarrow \sum_{1\le x\le p, 1\le y\le q} K_{p,q} F_{p,q} \quad (8).$$

The convolutional layer produces a two-dimensional array with dimensions $\frac{r-p+1}{k} \times \frac{v-q+1}{k}$ after applying several filters to the data. A three-dimensional array or a convolution layer with many filters can be the result. When padding is applied, the output often becomes longer but thinner to prevent it from becoming thinner. A quantum filter maps windows of form $p \times q$ into quantum states $\left|\psi^x\left(K_{f:(f+p-1),u:(u+q-1))}\right.\right\rangle$ of $Q = pq$ qubits by evolving the quantum state with a parametric quantum circuit $rc(\theta)$. The output quantum state $\left|\psi^o\right\rangle$ that results from this is

$$\left|\psi^0\right\rangle = rc(\theta)\left|\psi^x\left(K_{f:(f+p-1),u:(u+q-1))}\right.\right\rangle \quad (9).$$

Following the evolution, the expected value of the observable $Z^{\otimes Q}$ is found, enabling the feature map to be written as

$$K_{f:(f+p-1),u:(u+q-1))} \rightarrow \left\langle \psi^o \middle| Z^{\otimes Q} \middle| \psi^o \right\rangle \quad (10).$$

According to equation (10), nonlinearity in the quantum convolutional layer can be produced without the need for additional nonlinear functions, such as ReLU. It is appropriate for NISQ scenarios and experimentally accessible since it necessitates a minimum of qubits equal to the window size. The architecture keeps fully linked and pooling layers even if it lowers processing costs and might result in nonlinearities. The HQCCNN's loss function $G(\vec{\theta})$ can be expressed as follows:

$$G(\vec{\theta}) = g\left(\left\langle \psi^o \middle| Z^{\times Q} \middle| \psi^o \right\rangle\right) \quad (11).$$

In this case, g stands for a conventional function. Both the overall performance of the suggested approach and the convergence of the loss function are improved by optimizing the HQCCNN's loss function using the BSTO algorithm.

a. Boosted Sooty Tern Optimization

Onychoprion fuscatus, commonly referred to as sooty terns, is a species of seabird. As omnivores, they consume a wide range of creatures, including fish, amphibians, insects, reptiles, and earthworms. They are intelligent, colony organisms that find and hunt prey. Seasonally migrating in search of plentiful food supplies, sooty terns congregate to prevent collisions. For air attacks, they employ a flapping mode while in flight, adjusting their starting positions according to the sooty tern that is the fittest. The application of BSTO results in effective loss function minimization [16]. BSTO is used to minimize the loss function of HQCCNN. The fitness function is given in equation (12).

$$Fitness\ function = Min(Loss) \quad (12).$$

Where $Loss$ denotes the loss function of HQCCNN.

The proposed method involves data acquisition, preprocessing through correlation coefficients and normalization, and feature selection using a hybrid Osprey optimization algorithm and an enhancing sparrow search algorithm. The features are then fed into a quantum-classical convolutional neural network, optimized using boosted sooty tern optimization for attack detection.

IV. RESULTS AND DISCUSSION

In terms of accuracy, precision, and recall, the suggested approach performs better on the dataset than the traditional models. To improve stability and dependability, pre-processing, feature selection, and optimization procedures are combined. The method works well across the dataset and has greater detection accuracy.

A. Dataset Description

Three websites are used to gather ransomware traffic: Interactive Online Malware Analysis Sandbox, PacketTotal, and Malware-Traffic-Analysis.net. The following seven ransomware classes are protected by TLSv1.0 or TLSv1.2: GandCrab, Shade, Cryptowall, Cerber, WannaCry, Locky, and Teslacrypt. A normal sample is chosen at random from the CICIDS2017 Monday sub-dataset [17]. Table 1 shows the dataset description.

TABLE I DATASET DISTRIBUTION

Normal	20000
GandCrab	4172
Shade	1195
Cryptowall	1161

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Cerber	569
WannaCry	453
Locky	340
Teslacrypt	285

B. Comparison of performance with existing methods

The effectiveness of the suggested strategy is contrasted with other existing strategies in this section. The proposed method has been compared with well-known methods such as Xception ColSeq [11], EDVT-MSSAN [9], RF [10], and RANSOMNET+ [8]. The comparison uses several key performance metrics, including F1 score, specificity, recall, accuracy, and precision, to demonstrate the efficacy of the proposed method at detecting.

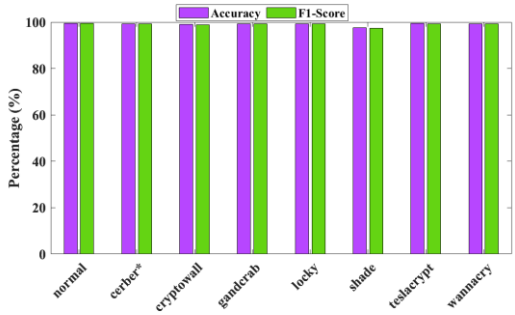

Figure.2. Accuracy and F1-Score Comparison

Figure 2 is a bar chart comparing the accuracy and F1 score of a classification model across different categories: Normal, GandCrab, Shade, Cryptowall, Cerber, WannaCry, Locky, and Teslacrypt. Every category has two bars, one containing the accuracy value (indicated in purple) and the second containing the F1-score value (indicated in green). Both of these values are above average and close to 100 percent, which shows that the model effectively diagnoses such categories. This is important when determining how suitable the model of the desired separation of data kinds in a ransomware attack.

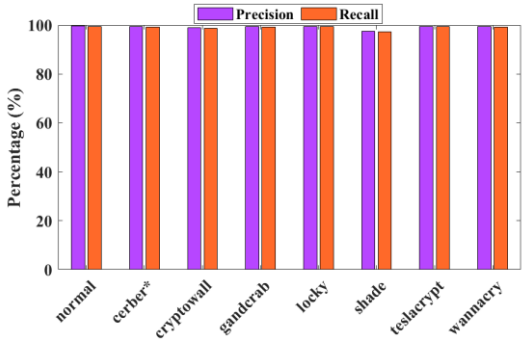

Figure.3. Precision and Recall Comparison

Figure 3 illustrates the precision and recall comparison of ransomware attacks. The x-axis lists the categories: Normal, GandCrab, Shade, Cryptowall, Cerber, WannaCry, Locky, and Teslacrypt. The y-axis means that the measure is a percentage, calculated to the nearest whole number ranging between 0 percent and 100 percent. Precision is represented in purple color, and recall is represented in orange color. Precision and recall show almost identical trends, both close to 100% in all categories, which suggests that the given detection method works well for every type of ransomware and normal activity.

TABLE II COMPARISON OF PERFORMANCE METRICS FOR DIFFERENT TECHNIQUES

Techniques	Accuracy (%)	Precision (%)	F1-Score (%)	Recall (%)	Specificity (%)
RANSOMNET+ [8]	88.8	87.5	87.8	88	89.2
EDVT-MSSAN [9]	90.1	88.5	89.4	89.8	89.5
RF [10]	87.1	85.6	85.8	87	88
Xception ColSeq [11]	86.1	84.8	85	85.1	86
Proposed HQCCNN-BSTO	99.7	99.5	99.5	99.4	99.4

Table 2 shows the effectiveness of various ransomware detection methods in terms of recall, specificity, accuracy, precision, and F1 score. RANSOMNET+, EDVT-MSSAN, RF, and Xception ColSeq are among the models that have been tested; their performance metrics range from moderate to high, with accuracies ranging from 85% to 90%. The suggested approach, HQCCNN-BSTO, on the other hand, performs noticeably better than these methods in every parameter, obtaining almost flawless results: 99.7% as determined by average accuracy and 99.5% as determined by precision and F1-score. This demonstrates that the suggested approach is appropriate for ransomware detection.

TABLE III COMPARISON OF ERROR RATE AND COMPUTATIONAL TIME FOR DIFFERENT METHODS

Methods	Error rate	Computational Time (s)
RANSOMNET+ [8]	1.78	1.8
EDVT-MSSAN [9]	1.56	1.6

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RF [10]	1.45	1.3
Xception ColSeq [11]	1.11	1.2
Proposed HQCCNN-BSTO	0.01	0.02

Table 3 presents the error rates and computer time of different approaches in ransomware detection. Computational results of RANSOMNET+, EDVT-MSSAN, RF, and Xception ColSeq have 1.11%-1.78% error rates, respectively, and the computational time varies from 1.2 to 1.8 sec. However, the proposed HQCCNN-BSTO approach performs much better, with a mean square error of 0.01%, and takes 0.02 sec, far less computational time than other methods. This means that the proposed method not only reduces the error rate but also offers functionality with significantly improved speed compared to other methods.

V. CONCLUSION

The proposed HQCCNN-BSTO approach proves to be very efficient and precise in the context of ransomware attack detection. By utilizing QCNN, added with BSTO, the proposed model demonstrates deterministic performance with an accuracy of 99.7% and an F1 score of 99.5%, which is far better than the existing techniques. The work also incorporates an enhanced preprocessing step employing correlation coefficients followed by min-max normalization, and the precise feature selection is based on the Hybrid Osprey Optimization Algorithm and the Enhancing Sparrow Search Algorithm to guarantee data robustness and feature relevancy. Furthermore, the proposed method includes an error rate of only 0.01%, while the computation time is very brief, 0.02 seconds. It also suggests that it can be used in real-time and also for the massive scope of cybersecurity applications. Overall, thus, the HQCCNN-BSTO method proves to be efficient in countering the new pattern of ransomware threats and provides a feasible and efficient solution for protecting the systems and data from severe ransomware attacks.

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